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Burden of disease modelling should be grounded in local knowledge

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Check for updates

Disease modelling for low-income settings often lacks reliable data, which leads to modelled outputs that contrast with empirical data and local knowledge



The global demand for annual reporting on key health indicators coupled with limited surveillance systems has fuelled the use of models to estimate disease burden[1–3]. These predictions are used by international agencies to inform global priorities, resource allocation and progress assessments towards health targets, including projections under scenarios such as funding gaps or climate change[4]. International health agencies and financing mechanisms also depend heavily on modelled outputs to illustrate policy impacts and justify investments.

This reliance overlooks the fact that the underlying data may be weak or, in some instances, completely absent. In many low-resource environments, the lack of strong civil registration and vital statistics (CRVS) leads to dependence on outdated, sparse and unevenly distributed data. Many modelled predictions present associated estimates of uncertainty that reflect the outputs generated under specific assumptions but rarely present the uncertainty that underlies those assumptions. Visually striking outputs, such as high-resolution disease maps, can hide the fragility of their empirical bases. For instance, the Malaria Atlas Project and the Global Burden of Disease (GBD) have estimated malaria burden across sub-Saharan Africa using about 66,000 malaria prevalence survey points to interpolate predictions in over 955,000 5 × 5-km grid squares annually since 2000. These models translate predicted prevalence data into incidence and mortality estimates using layered assumptions, including climate correlations and limited case-fatality data. Much of this is derived from fewer than 100 verbal autopsy studies conducted in sub-Saharan Africa between 2000 and 2017, a method known to be imprecise for malaria. More recently, minimal invasive tissue sampling was used in a few highly specific sites during 2018–2019, which further constrains the generalizability of the findings[2]. Although the figures are frequently cited, the validity of the underlying assumptions and data scarcity is rarely examined.

Responsible prediction

Falsely assumed precision can mislead policymakers and funders, as evidenced by early COVID-19 models that predicted unrealistic outcomes due to flawed assumptions[5,6]. Many of these models are created in academic or consultancy settings with limited direct input from the actual contexts they aim to represent. The gap between the modelling industry and local systems, stakeholders and epidemiologists is not just practical but also epistemic[7]. When modellers work without understanding field realities, their results can lack relevance or even mislead intervention efforts. As Byass warned over a decade ago, global models that combine data from high-income areas with limited data from low-income regions often suffer from limited local relevance and insufficient validation[8], as illustrated when COVID-19 case fatality figures were generalized to Africa (table 1 of ref. 6). Nevertheless, these models continue to influence worldwide health funding and public health policy.

In 2022, Murray indicated that "The GBD estimates each quantity of interest for every location. Even when data are highly inconsistent or there are no data for a disease or risk, a best estimate is produced along with our best estimate of uncertainty. The logic is that decisions have to be made, and a best estimate borrowing insight from where data are available is better than no estimate, provided that there is clarity around the level of uncertainty"[9]. Although models may offer valuable possible outcomes (especially during crises), this benefit depends on clearly communicating their limitations, and their use in crisis may increase the adverse consequences of incorrect decisions. Overconfident or poorly supported estimates can cause more harm than good, especially when used for resource distribution or prioritizing interventions[5]. All models imperfectly reflect reality. Their strength lies in quantifying uncertainty through careful choices of model structure, priors and calibration. Methodological advances are increasing in complexity, and combining mechanistic, statistical and geospatial modelling, often referred to as ensemble models or bespoke models; however, they cannot overcome fundamental issues such as sparse data or lacking local context (discussed in 'A call for data-driven public health' below). Without validation and methodological transparency, complex models risk becoming 'black boxes', that project sophistication while obscuring their weaknesses. Models that lack local context can also be misleading or inaccurate, as they often rely on flawed or overgeneralized assumptions. When data sources and sampling methods are poorly understood or far removed from their origin, uncertainty estimates can become layered and difficult to interpret.

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Countries often first see their national estimates in globally produced disease estimates upon publication, which hampers their ability to contextualize or challenge these figures using local data and knowledge, or to understand the models and uncertainties before making a public response. The global pool of modelling expertise is limited, and as methods grow more complex there is a pressing need to standardize and communicate the underlying assumptions and data inputs effectively to end users. Without this, there is a risk of fostering a scientific culture in which only a small group of technical experts understand the methods, assumptions and uncertainties involved. Consequently, policymakers at both regional and national levels may be unable to make reasoned decisions based on model outputs because they are impossible to interpret.

A call for data-driven public health

The foundation of effective public health policy relies on timely, accurate and locally relevant data. Many low- and middle-income countries, especially across sub-Saharan Africa, have weak health data infrastructures, which makes them rely more on modelled estimates than empirical data[2,6]. Moving towards data-driven public health requires long-term, strategic efforts to develop robust national data ecosystems and shift research priorities towards surveillance, validation and system resilience[10].

Primary data collection is essential for responsible health decisions and should not be regarded as just a methodological choice or as a less cost-effective alternative to modelled predictions. Predictions from a model should be referred to as predictions, not as 'data'. Systems such as civil registration and vital statistics and national health information systems are crucial pillars of evidence-based health frameworks. The key lesson is that sustainable health intelligence cannot be outsourced. Progress depends on countries taking the lead in designing, funding and managing their data systems. This national ownership involves establishing data agendas aligned with local priorities, developing analytical capacities within the country, and maintaining control over data governance and usage[11,12]. Donors and international agencies need to encourage a shift away from dependence on external expertise to support, by focusing on empowering local efforts and rejecting generic technical solutions. Efforts to address weak data by using more complex estimation methods is not the solution.

Addressing these systemic problems will inevitably require long-term investment and deeper structural change. However, there are some immediate investments that could begin to validate, calibrate and ultimately replace disease burden models (Box 1).

Data are an essential strategic asset. Without reliable data, health systems remain reactive, inefficient and inequitable. Proper investment enables countries to plan, allocate and act with precision and foresight, to support health equity and evidence-based public health.

Ways forward

There is a clear need to improve the quality, quantity and visibility of local data collection across low- and middle-income countries. However, this is at variance with the funding invested in large, external modelling groups and the limited funding invested in strengthening field data systems. It is worth noting that 2025 marks 20 years since the launch of the Health Metrics Network, which aimed to strengthen health information systems in low-resource settings — goals that are being revisited today. Despite initial momentum, the Health Metrics Network was discontinued in 2013, while initiatives funding modelling consortia have expanded. This raises important questions about progress and priorities in global health data[13]. For example, there are only a handful of sites across Africa that have systematically collected longitudinal data on malaria disease over 20 years since the launch of the Roll Back Malaria Initiative in 2000. Reliable estimates will come from a collaborative effort among data collectors, public health experts and modellers from the ground up, rather than from top-down extrapolations.

Models should not surpass or replace the data they aim to synthesize and represent. Emerging research advocates for a participatory model of development, which stresses the importance of engaging local stakeholders, using open data and scrutinizing assumptions[14]. These practices enhance model accuracy, foster local ownership, improve interpretability and align better with ground-level decision making.

Investments in improving disease surveillance, including a combination of gold-standard sentinel sites and broader low-cost surveillance at scale, should be prioritized alongside funding to improve modelling

BOX 1

Investments to improve and replace disease burden models

Validation research
Locally led analyses that compare modelled predictions with real-world data to understand where models succeed, where they fail and how they can be improved. This can enable an assessment of whether model predictions of rising, stagnating or falling disease burdens are correct across settings, or whether a given disease exists where models predict its existence or non-existence.

Establish country-led surveillance sites
Fund strategically placed sites that act as sentinel sites to strengthen national capacity for disease risk assessment. For example, minimal investment in hospital surveillance can markedly improve information on local causes of severe illness and death, as well as how these change when new interventions (such as vaccines, drugs and community care) are introduced[15].

Reduce reliance on external expertise and funding
Build local capacity by training health officials in informatics, epidemiology and data analytics; this is fundamental for a future in which local data and models can inform local disease policy.

Improve investment in foundational data systems that support civil registration and vital statistics and health information systems
Enhance these systems through digital integration, mobile reporting tools and routine community surveillance; this will not only improve the quality and availability of local data but also empower local actors with the skills and tools necessary to effectively analyse and use that data for informed decision making.

Financial stability
Develop sustainable data funding strategies, domestic and blended, that protect data systems from donor volatility and strengthen national ownership.

of incomplete data. Without strengthening surveillance, modelling will remain a workaround instead of a complementary tool.

Academic journals have fostered an acceptance of global disease modelling over granular observational data. There is an epistemic element to this in which computation and mathematics providing global estimates without local context is considered higher value than on-the-ground surveillance in a specific context. Journal editors should be encouraged to accept and promote the value of long-term epidemiological and country-specific evidence in peer-reviewed journals.

Conclusion

This Comment is not intended to oppose disease modelling or its value in public health analytics but to provide a context for its appropriate use and future application, and to emphasize the foundational importance of real data. To regain credibility and usefulness, a fundamental change is needed in how modelling is used, introduced and understood in environments with scarce data.

Responsible prediction starts with recognizing data limitations and advocating for improved data, and then transparently calculating the uncertainty regarding extrapolations. It involves communication of uncertainty (including the uncertainty derived from model fit and from assumptions), rigorous validation with local data sources, and meaningful engagement with local experts[7].

Acknowledging the limits of modelling in data-scarce environments is a call for scientific integrity. The credibility of global health models relies not only on technical expertise but also on ethical and epistemological responsibility. Moving forward, modelling in data-limited settings should focus on generating disease estimates thoughtfully, grounded in local knowledge and validated where possible using local data, and aligned with the needs of countries to know what the 'real situation' is rather than global donor communities reporting needs.

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Competing interest

The authors declare no competing interests.